

AI Candidate Ranking System

E2E Pitch Deck & Evaluation Report

Problem Statement

- Traditional ATS filters rely on simple keyword matching and are easily gamed.
- Candidates engage in keyword-stuffing (e.g. listing Python/AI skills with no experience).
- Mismatches exist between candidate titles and actual career description narratives.
- Scalability issues: processing 100,000+ profiles manually or with heavy LLMs is impossible.

System Architecture

1. Data Ingestion: Stream JSONL to handle large 487MB dataset in constant memory.
2. Fast Semantic Filter: Compute lightweight Jaccard/Sentence Embeddings on all candidates.
3. Deep Scorer: Calculate precise skill fit weights, verification penalties & behavioral signals.
4. LLM Evaluator: Batch calls to DeepSeek API (with rate limiters) for narrative reasonings.
5. Fallback Mode: Seamless transition to rule-based templates if DeepSeek is offline.

Scoring Methodology

Final Score = $0.40 * \text{Technical} + 0.35 * \text{Career} + 0.25 * \text{Behavioral}$

- Technical (40%): Semantic similarity, required skill overlap, core tools.
- Career (35%): Experience (target 5-9, peak 6-8), current title, product company history.
- Behavioral (25%): Login recency, open to work status, response rate, notice period.
- Hard Filters & Honeypots: Auto 0.0 score (consulting-only, academic, impossible YoE).

Key Data Insights

- Total candidates processed: 100
- Average candidate composite score: 0.4862
- Highly endorsed skills: Python, PyTorch, SQL.
- Notice period penalty successfully applied to long notice profiles.

Top 5 Ranked Candidates

Candidate ID	Composite Score	Reasoning Summary
CAND_0053605	0.6204	We highlight this profile due to strong career matching. The c...
CAND_0008295	0.5838	A solid candidate with a background as a AI Research Engineer ...
CAND_0010149	0.5794	Demonstrating exceptional career fit, this ML Engineer possess...
CAND_0054859	0.5628	Demonstrating exceptional career fit, this Senior Software Eng...
CAND_0080766	0.5472	We highlight this profile due to strong career matching. The c...